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Approaches in a sensor model of error correction in dynamic measurements

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Abstract

The article describes three approaches in error correction of dynamic measurements. A sensor model is used in these approaches. The first one is based on the modal control of the dynamic behavior and adapting parameters of a sensor model by direct search. We propose a method of error evaluation in dynamic measurements with a priori information on characteristics of measured signal and noise of the sensor available. The second approach concerns the neural network representation of a sensor. Neural network inverse sensor model and the algorithm for its training by minimizing mean-squared dynamic measurements error criterion are proposed. The third approach is based on the introduction of sliding mode control into the measuring system with modal control of dynamic behavior to achieve the similarity of the sensor model output to the sensor output.

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Keywords: dynamic measurements error; model of sensor; recovery of sensor input signal; dynamic measuring system; automatic control theory approach; sliding mode control approach; neural networks approach.

1. Introduction

Correction of the dynamic measurements error consists in the solution to problem of a measuring system input signal recovery. This problem is ill-posed and requires the numerical solution to the convolution integral equation [1]–[5]. One of the promising tendencies in the area of modern information-and-measuring equipment development is an intellectualization of such equipment [6]. Characteristic features thereof are the completion of complex measuring procedures using dedicated equipment and means, as well as the development of detectors capable of

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individualizing the processing algorithms, including by adapting change of its own structure and parameters based on accumulated a priori and obtained measuring information [7]. In real information-measuring systems noise characteristics are approximated and may change during measurement. This makes it difficult to process the results of measurement in appropriate way. That is why intellectual measuring systems, adaptable to noise environment, are of practical interest. Discussed in studies [8]–[16] detector with sensor model, existing as real part of its structure, had the model feedback loop coefficients acting as adjustable parameters. Such coefficients was calculated according to the required dynamic error value and remained constant during the experiment data processing. However, there is a possibility in principle to adjust the sensor model feedback loop coefficients during measurement or measurement data processing [17], [18]. In this case the parameter adjustment criterion may serve not only as dynamic error convex functional, but also as dedicated generated signal of dynamic error evaluation, which is a continuous time function, depending on adjustable parameters of the measuring system.

The automatic control theory approach allows the effective improvement of the dynamic measurements accuracy [15], [16]. Along with it, the artificial neural network approach to creation of dynamic models of measuring systems and algorithms for data processing of dynamic measurements is one of promising ways of intelligent measuring systems development.

2. Dynamic behavior modal control approach

2.1. Gradient method of the measuring system dynamic parameters minor adjustment

Let's consider a common case of the measuring system dynamic parameters self-adjustment, when all feedback loops coefficients are subjected to minor adjustment. The method is intended for adjusting the parameters, when the spectral noise density of the measuring channel is constant and unknown, or is slowly changing with a speed not exceeding the speed of the system transient processes.

Self-adjustment criterion of the measuring system dynamic parameters shall adequately reflect noise conversion in the measuring channel, in compliance with a transfer function according to reduced noise term from [15]. The evaluation of the measuring system dynamic error shall be written as follows [16]:

$$e_0(t) = y(t) + V(t) - y_m(t) \quad (1)$$

where $y(t)$ is input signal of the measurement system; $V(t)$ is signal to noise; $y_m(t)$ is input signal of the sensor model.

Or using Laplace transformation, considering the transfer functions of the sensor and the measuring system:

$$e_0(p) = e_{ms}(p)W_D(p) + V(p) \quad (2)$$

where $e_{ms}(p) = U(p) - U^*(p)$ is a Laplace transformation of the measuring system error; $W_D(p)$ is Laplace transformation of the sensor's transfer function.

A signal given below shall be taken for the displacement error of the measuring system and the model output coordinates to self-adjustment criteria:

$$e_1(p) = e_0(p) \frac{W_{ms}(p)}{W_D(p)} = e_{ms}(p) \cdot W_{ms}(p) + V(p) \frac{W_{ms}(p)}{W_D(p)} \quad (3)$$

On condition that $W_{ms}(p) \approx 1$ the signal ensures noise conversion in the error signal with the same degree as its amplification in the output signal of the measuring system. Measure of disagreement of the real system output coordinate motion and the model output shall be the function from the dynamic error evaluation signal – $e_1^*(t)$, in the form as follows [15]:

$$I(t) = F(e_1^*(t)) = (e_1^*(t))^2 \quad (4)$$

where $I(t)$ is a self-adjustment quality criterion, depending on the input signal – $U(t)$, initial displacement of the model and system coordinates, as well as on adjustable dynamic parameters $k_i(t)$, $i=0, n-1$.

Algorithm of small adjustment of the measuring system dynamic parameters $k_i(t)$, $i=0, n-1$ in compliance with the gradient method shall be developed in such manner, that at every instant the feedback loop coefficient change tends to $I(t)$ value decrease as the function $k_i(t)$:

$$\frac{dk_i(t)}{dt} = -\lambda \cdot \frac{\partial I(t)}{\partial k_i(t)} \quad (5)$$

The analysis performed based on Lyapunov's method [15] has demonstrated that the stability of the measuring system, synthesized using a gradient method, is determined by additional feedback loops with coefficients $x_{nm}^0(t), x_{n-1m}^0(t), \dots, x_{1m}^0(t)$, which index depends on the system motion parameters. Value of additional feedback loops is proportional to the adaption speed λ , which control gives possibility of ensuring a measuring system stability.

2.2. Results of computer simulation

To illustrate the possibilities of the dynamic error correction a computer simulation of measuring systems with dynamic error self-adjusting feedback loops coefficients of the sensor model, developed based on the first-order (fig. 1) and the second order sensors was performed. The results of the computer simulation of the dynamic measurement process demonstrate the effectiveness of the designed self-adjusting measuring system. In case of self-adjusting of parameters of the measuring system based on the second sensor with time constant $T = 0.01$ and damping ratio $\xi = 0.3$, with harmonic input signal – $U(t) = 1 \cdot \sin(10 \cdot t)$ and harmonic noise signal – $V(t) = 0.05 \cdot \sin(100 \cdot t)$, dynamic measurement error has reduced by 84% compared to measurement without additional adjustment.

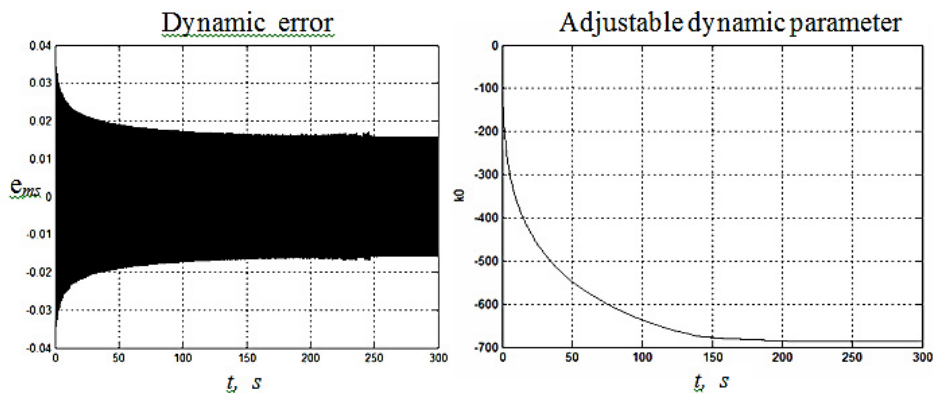


Fig.1. Results of the computer simulation

2.3. Summary

The proposed structure of the measuring system with modal dynamic characteristics control based on sensor model dynamic error and dynamic error assessment channel, as well as self-adjustment algorithm of the measuring system dynamic parameters, based on gradient method, allow to decrease dynamic measurement error by changing

adjustable parameters directly in the process of measurement or at the stage of the measuring experiment data processing.

3. Neural network approach

3.1. Neural network inverse model of a sensor

Let a primary measuring transducer (sensor) is described by the transfer function (TF) as follows:

$$W_s(p) = \frac{Y(p)}{U(p)} = K_0 \frac{\prod_{i=1}^k (T_{2i}^2 p^2 + 2\xi_{2i} T_{2i} p + 1) \prod_{i=k+1}^m (T_{2i} p + 1)}{\prod_{j=1}^q (T_{1j}^2 p^2 + 2\xi_{1j} T_{1j} p + 1) \prod_{j=q+1}^n (T_{1j} p + 1)} \quad (6)$$

where U and Y are the sensor input and output signals respectively; T_{1j} and T_{2i} are time constants; ξ_{1j} and ξ_{2i} are damping coefficients; $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$ ($m \leq n$); K_0 is the static gain; p is the complex number frequency.

The recovery of the input signal of the sensor with TF (6) is carried out by its measured output signal processing on a basis of the sensor inverse model. This model is presented as the sequential connection of the correcting filter and identical first-order sections, each of which is the neural network inverse model of the following first-order TF:

$$W_1(p) = \frac{1}{pT_1 + 1} \quad (7)$$

The value of the time constant T_1 in the TF (7) is set equal to such a value among time constants T_{1j} in the TF (6), that provides the proximity of step responses of systems with TFs (6) and (7). The TF $W_{cf}(p)$ of the correcting filter is the inverse TF of the sensor, which is supplemented with a certain number ($n_2 - m_2$) of TFs (1) to ensure the stability of the inverse model.

The recovery of the dynamically distorted input signal of the sensor on the basis of its neural network inverse model can be accompanied by the significant increase of the additive noise at the sensor output, as well as the internal noise of the inverse model. For the correct recovery of the input signal of the sensor it is expedient to expand the neural network inverse model, taking into account the presence of the additive noise at the sensor output. This expansion can be implemented as the additional low-pass filtration of the recovered signal by means of the increase of the order of sequential neural network sections in the structure of the inverse model. The block diagram of the d -order section is shown in fig. 2. The algorithm of training sets generation and their length evaluation for the section $C_d[T_1]$ parameters adjustment during the training procedure was proposed [19], [20].

3.2. Results of the experimental data processing

The algorithm for the recovery of dynamically distorted signals on a basis of the proposed neural network inverse model of the sensor was developed. In order to validate experimentally the efficiency of these model and algorithm the dynamic measurement of the temperature was made. The step response of the thermoelectric transducer (thermocouple) «Metran-281» by heating it from 0 °C to 800 °C was obtained.

The result of experimental data processing at $d = 66$ in the form of the plots of the thermocouple measured output $y(t)$ and the thermocouple recovered input $u^*(t)$ is shown in fig. 3.

The obtained result shows that the time of the dynamic temperature measurement decreased from $T_s = 306$ s to $T_d = 60$ s, that is more than 5 times. This validates the efficiency and the effectiveness of the proposed model and the algorithm of the dynamic measurements error correction.

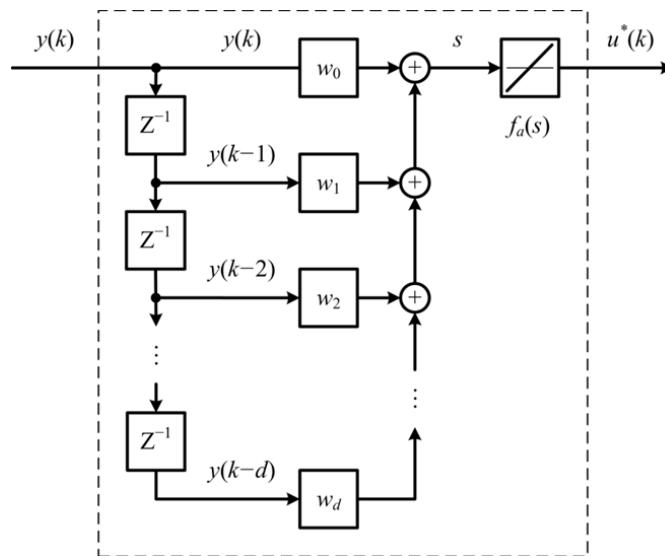
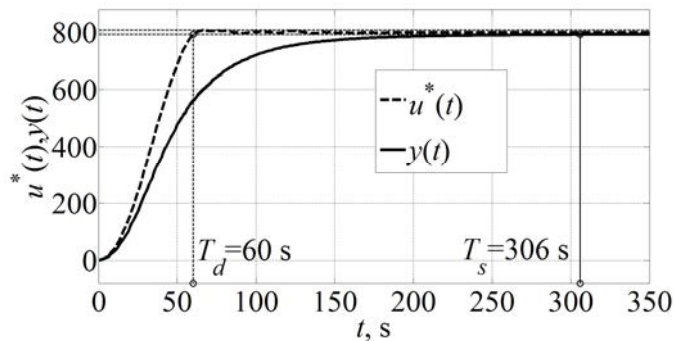
Fig. 2. Block diagram of the d -order section

Fig.3. Result of experimental data processing

3.3. Summary

The neural network approach to the recovery of dynamically distorted signals allows the effective correction of the dynamic measurements error caused by the inertia of the sensor and the additive noise at its output.

The considered neural network inverse model of a sensor with filtration of sequentially recovered signal allows the effective improvement of the sensor dynamic behaviour due to deep mathematical processing of measurement data.

The result of experimental data processing validates the improvement of the sensor dynamic behaviour due to the application of the proposed model and the algorithm to the dynamic measurement of the temperature.

4. Conclusions

The proposed structure of the measuring system with modal dynamic characteristics control based on sensor model dynamic error and dynamic error assessment channel, as well as self-adjustment criterion of the measuring system dynamic parameters, based on gradient method, allow to decrease dynamic measurement error by changing

adjustable parameters directly in the process of measurement or at the stage of the measuring experiment data processing.

The neural network approach to the recovery of dynamically distorted signals allows the effective correction of the dynamic measurements error caused by the inertia of the sensor and the additive noise at its output. The considered neural network inverse model of a sensor with filtration of sequentially recovered signal allows the effective improvement of the sensor dynamic behaviour due to deep mathematical processing of measurement data.

Moreover, to ensure the proximity of the sensor model output to the sensor output in the measuring system with modal control of dynamic behavior feedbacks are introduced. It is possible to achieve the proximity of these signals in the measuring system by implementation of sliding mode control [21], [22].

The results of computer modeling and experimental data processing validate the improvement of the sensor dynamic behaviour due to the application of proposed models and algorithms to the dynamic measurements error correction.

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